EM waves

$$\frac{\partial F^{\alpha\beta}}{\partial x^\beta} = -\mu_0 j^\alpha = 0$$ in the absence of any charges (that is, in vacuum).

$$\Rightarrow \frac{\partial}{\partial x^\beta}\left(\frac{\partial A^\beta}{\partial x_\alpha} - \frac{\partial A^\alpha}{\partial x_\beta}\right) = 0$$

$$\Rightarrow (1)\ \frac{\partial^2 A^\beta}{\partial x_\alpha \partial x^\beta} - \frac{\partial^2 A^\alpha}{\partial x_\beta \partial x^\beta} = 0$$

Note: β is repeated $\rightarrow$ summed over. $\alpha = 0, 1, 2, 3 \Rightarrow$ 4 equations in total.

But there is gauge freedom: can add $\frac{\partial g}{\partial x_\alpha}$ to form

$$\tilde{A}^\alpha = A^\alpha + \frac{\partial g}{\partial x_\alpha}$$. This does not change $\vec{E}, \vec{B}$

and $$\tilde{F}^{\alpha\beta} = \frac{\partial \tilde{A}^\beta}{\partial x_\alpha} - \frac{\partial \tilde{A}^\alpha}{\partial x_\beta} = \frac{\partial}{\partial x_\alpha}\left(A^\beta + \frac{\partial g}{\partial x_\beta}\right) -$$

$$- \frac{\partial}{\partial x_\beta}\left(A^\alpha + \frac{\partial g}{\partial x_\alpha}\right) = \underbrace{\frac{\partial A^\beta}{\partial x_\alpha} - \frac{\partial A^\alpha}{\partial x_\beta}}_{F^{\alpha\beta}} + \frac{\partial^2 g}{\partial x_\alpha \partial x_\beta} - \frac{\partial^2 g}{\partial x_\beta \partial x_\alpha}$$

$$\Rightarrow \tilde{F}^{\alpha\beta} = F^{\alpha\beta}.$$

Thus, $F^{\alpha\beta}$ does not change (is invariant under the gauge transformation)

therefore $\frac{\partial}{\partial x^\beta} F^{\alpha\beta} = 0$ also does not change.

We can use such a gauge freedom to simplify Eq.(1).

Namely, we can always choose A^α to satisfy <u>Lorentz gauge</u>

$$\frac{\partial A^\alpha}{\partial x^\alpha} = 0.$$

If so, then (1) turns into $\dfrac{\partial^2 A^\alpha}{\partial x_\beta \partial x^\beta} = 0$

$$\Rightarrow \frac{1}{c^2}\frac{\partial^2 A^\alpha}{\partial t^2} - \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right) A^\alpha = 0$$

or, $\dfrac{1}{c^2}\dfrac{\partial^2 A^\alpha}{\partial t^2} - \nabla^2 A^\alpha = 0$

or, $\Box = \nabla^2 - \dfrac{1}{c^2}\dfrac{\partial^2}{\partial t^2}$ (D'Alembertian)

$\Box A^\alpha = 0$. This holds for every α.

<u>Clarification</u>: how can we make sure that $\dfrac{\partial A^\alpha}{\partial x^\alpha} = 0$ holds?

Suppose we have A^α such that $\dfrac{\partial A^\alpha}{\partial x^\alpha} \neq 0$.

Then we introduce $\tilde{A}^\alpha = A^\alpha + \dfrac{\partial g}{\partial x_\alpha}$ and require that $\dfrac{\partial \tilde{A}^\alpha}{\partial x^\alpha} = 0$

$\Rightarrow$ but $\dfrac{\partial \tilde{A}^\alpha}{\partial x^\alpha} = \dfrac{\partial A^\alpha}{\partial x^\alpha} + \dfrac{\partial^2 g}{\partial x^\alpha \partial x_\alpha}$

$\Rightarrow$ we obtain differential equation on function

$$g: \quad \frac{\partial^2 g}{\partial x_\alpha \partial x^\alpha} = -\frac{\partial A^\alpha}{\partial x^\alpha}.$$

Explicitly, $\underbrace{\nabla^2 g - \dfrac{1}{c^2}\dfrac{\partial^2 g}{\partial t^2} = \dfrac{\partial A^\alpha}{\partial x^\alpha}}$.

$g(\vec{r}, t)$ must satisfy

Then $\tilde{A}^\alpha$ will satisfy $\dfrac{\partial \tilde{A}^\alpha}{\partial x^\alpha} = 0$ and $\Box \tilde{A}^\alpha = 0$.

Consider now a <u>plane wave</u> solution of $\Box A^\alpha = 0$.

let $A^\alpha(\vec{r},t) = A_0^\alpha f(x-vt)$

$\hookrightarrow$ 4-vector amplitude (constant).

$$\Rightarrow \quad \nabla^2 A^\alpha(\vec{r},t) = A_0^\alpha \frac{\partial^2}{\partial x^2} f(x-vt) = A_0^\alpha f''(x-vt)$$

$$\frac{\partial^2 A^\alpha(\vec{r},t)}{\partial t^2} = A_0^\alpha \frac{\partial^2}{\partial t^2} f(x-vt) = A_0^\alpha f''(x-vt) \cdot \left(\frac{\partial}{\partial t}(x-vt)\right)^2.$$

$$\Rightarrow \quad \left(\nabla^2 - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\right) A^\alpha(\vec{r},t) = 0 \quad \text{gives}$$

$$A_0^\alpha \left(1 - \frac{v^2}{c^2}\right) f''(x-vt) = 0.$$

This is satisfied by any non-zero function f provided that $v^2 = c^2 \Rightarrow v = \pm c$.

$\Rightarrow$ for $v = +c$ $f(x-vt) = f(x-ct)$ describes right-moving wave.

$v = -c$ $f(x-vt) = f(x+ct)$ — left-moving wave.

All waves propagate with the speed of light c.

Summary of Maxwell's equations

SI	CGS
$m\frac{du^\alpha}{d\tau} = q F^{\alpha\beta} u_\beta$	$m\frac{du^\alpha}{d\tau} = \frac{q}{c} F^{\alpha\beta} u_\beta$
$\frac{d\vec{p}}{dt} = \vec{F} = q\vec{E} + q\vec{v}\times\vec{B}$	$\frac{d\vec{p}}{dt} = \vec{F} = q\vec{E} + \frac{q}{c}\vec{v}\times\vec{B}$
$F^{\alpha\beta}$	$F^{\alpha\beta}\left(\text{with } \frac{\vec{E}}{c} \to \vec{E}\right)$
$\frac{\partial F^{\alpha\beta}}{\partial x_\beta} = -\mu_0 j^\alpha$	$\frac{\partial F^{\alpha\beta}}{\partial x^\beta} = -\frac{4\pi}{c} j^\alpha$

$$\frac{\partial F^{\alpha\beta}}{\partial x_\mu} + \frac{\partial F^{\mu\alpha}}{\partial x_\beta} + \frac{\partial F^{\beta\mu}}{\partial x_\alpha} = 0$$

$\vec{\nabla}\times\vec{E} = -\frac{\partial\vec{B}}{\partial t}$	$\vec{\nabla}\times\vec{E} = -\frac{1}{c}\frac{\partial\vec{B}}{\partial t}$
$\vec{\nabla}\cdot\vec{E} = \frac{\rho}{\varepsilon_0}$	$\vec{\nabla}\cdot\vec{E} = 4\pi\rho$

$$\vec{\nabla}\cdot\vec{B} = 0$$

$\vec{\nabla}\times\vec{B} = \mu_0\vec{j} + \frac{1}{c^2}\frac{\partial\vec{E}}{\partial t}$	$\vec{\nabla}\times\vec{B} = \frac{4\pi}{c}\vec{j} + \frac{1}{c}\frac{\partial\vec{E}}{\partial t}$